N1 Discrete probability distributions

Name:

Class:

Date:

Time: **371 minutes**

Marks: **304 marks**

Comments:

Q1.

Plastic clothes pegs are made in various colours.

The number of red pegs may be modelled by a binomial distribution with parameter p equal to 0.2.

The contents of packets of 50 pegs of mixed colours may be considered to be random samples.

(a) Determine the probability that a packet contains:

(i) less than or equal to 15 red pegs;

(2)

(ii) exactly 10 red pegs;

(2)

(iii) more than 5 but fewer than 15 red pegs.

(3)

(b) Sly, a student, claims to have counted the number of red pegs in each of 100 packets of 50 pegs. From his results the following values are calculated.

Mean number of red pegs per packet = 10.5

Variance of number of red pegs per packet = 20.41

Comment on the validity of Sly's claim.

(4)

(Total 11 marks)

Q2.

Kirk and Les regularly play each other at darts.

(a) The probability that Kirk wins any game is 0.3, and the outcome of each game is independent of the outcome of every other game.

Find the probability that, in a match of 15 games, Kirk wins:

(i) fewer than half of the games;

(4)

(ii) more than 2 but fewer than 7 games.

(3)

(b) Kirk attends darts coaching sessions for three months. He then claims that he has a probability of 0.4 of winning any game, and that the outcome of each game is independent of the outcome of every other game.

(i) Assuming this claim to be true, calculate the mean and standard deviation for the number of games won by Kirk in a match of 15 games.

(3)

(ii) To assess Kirk's claim, Les keeps a record of the number of games won by Kirk in a series of 10 matches, each of 15 games, with the following results:

8 5 6 3 9 12 4 2 6 5

Calculate the mean and standard deviation of these values.

(2)

(iii) Hence comment on the validity of Kirk's claim.

(3)
(Total 15 marks)

Q3.

Kirk and Les regularly play each other at darts.

(a) The probability that Kirk wins any game is 0.3, and the outcome of each game is independent of the outcome of every other game.

Find the probability that, in a match of 15 games, Kirk wins:

(i) exactly 5 games;

(3)

(i) fewer than half of the games;

(3)

(ii) more than 2 but fewer than 7 games.

(3)

(b) Kirk attends darts coaching sessions for three months. He then claims that he has a probability of 0.4 of winning any game, and that the outcome of each game is independent of the outcome of every other game.

(i) Assuming this claim to be true, calculate the mean and standard deviation for the number of games won by Kirk in a match of 15 games.

(3)

(ii) To assess Kirk's claim, Les keeps a record of the number of games won by Kirk in a series of 10 matches, each of 15 games, with the following results:

8 5 6 3 9 12 4 2 6 5

Calculate the mean and standard deviation of these values.

(2)

(iii) Hence comment on the validity of Kirk's claim.

(3)
(Total 17 marks)

Q4.

An express coach travels daily between Middlesbrough and London. The coach calls at Thirsk only when seats have been reserved in advance. On any day, the probability that the coach calls at Thirsk is 0.5, and is independent of that on any other day.

(a) Determine the probability that, during a 14-day period, the coach calls at Thirsk:

(i) on at most 10 days;

(ii) on more than 5 days but fewer than 10 days.

(5)

(b) When the coach calls at Thirsk, it is possible to purchase a seat on the coach providing seats are available. The probability that the coach calls at Thirsk with at least one seat available on any day is 0.4, and is independent of that on any other day.

Calculate the probability that, during a 28-day period, the coach calls at Thirsk with at least one seat available on exactly 7 days.

(3)

(c) Indicate why a single binomial model would **not** be appropriate for the number of calls **per month** of the coach at Thirsk.

(1)

(Total 9 marks)

Q5.

Each evening Aaron sets his alarm for 7 am. He believes that the probability that he wakes before his alarm rings each morning is 0.4, and is independent from morning to morning.

(a) Assuming that Aaron's belief is correct, determine the probability that, during a week (7 mornings), he wakes before his alarm rings:

(i) on 2 or fewer mornings;

(ii) on more than 1 but fewer than 5 mornings.

(5)

(b) Assuming that Aaron's belief is correct, calculate the probability that, during a 4-week period, he wakes before his alarm rings on exactly 7 mornings.

(3)

(c) Assuming that Aaron's belief is correct, calculate values for the mean and standard deviation of the number of mornings in a week when Aaron wakes before his alarm rings.

(2)

(d) During a 50-week period, Aaron records, each week, the number of mornings on which he wakes before his alarm rings. The results are as follows.

Number of mornings	0	1	2	3	4	5	6	7

Frequency	10	8	7	7	5	5	4	4

(i) Calculate the mean and standard deviation of these data.

(3)

(ii) State, giving reasons, whether your answers to part (d)(i) support Aaron's belief that the probability that he wakes before his alarm rings each morning is 0.4, and is independent from morning to morning.

(3)

(Total 16 marks)

Q6.

(a) At a particular checkout in a supermarket, the probability that the barcode reader fails to read the barcode first time on any item is 0.07, and is independent from item to item.

(i) Calculate the probability that, from a shopping trolley containing 17 items, the reader fails to read the barcode first time on exactly 2 of the items.

(3)

(ii) Determine the probability that, from a shopping trolley containing 50 items, the reader fails to read the barcode first time on at most 5 of the items.

(2)

(b) At another checkout in the supermarket, the probability that a faulty barcode reader fails to read the barcode first time on any item is 0.55, and is independent from item to item.

Determine the probability that, from a shopping trolley containing 50 items, this reader fails to read the barcode first time on at least 30 of the items.

(3)

(Total 8 marks)

Q7.

(a) At a particular checkout in a supermarket, the probability that the barcode reader fails to read the barcode first time on any item is 0.07, and is independent from item to item.

(i) Calculate the probability that, from a shopping trolley containing 17 items, the reader fails to read the barcode first time on exactly 2 of the items.

(3)

(ii) Determine the probability that, from a shopping trolley containing 50 items, the reader fails to read the barcode first time on at most 5 of the items.

(2)

(b) At another checkout in the supermarket, the probability that a faulty barcode reader fails to read the barcode first time on any item is 0.55, and is independent from item to item.

Determine the probability that, from a shopping trolley containing 50 items, this reader fails to read the barcode first time on at least 30 of the items.

(3)

(c) At a third checkout in the supermarket, a record is kept of X, the number of times per 50 items that the barcode reader fails to read a barcode first time. An analysis of the records gives a mean of 10 and a standard deviation of 6.8.

(i) Estimate p, the probability that the barcode reader fails to read a barcode first time.

(1)

(ii) Using your estimate of p and assuming that X can be modelled by a binomial distribution, estimate the standard deviation of X.

(2)

(iii) Hence comment on the assumption that X can be modelled by a binomial distribution.

(2)

(Total 13 marks)

Q8.

The probability that Janet receives at least one e-mail message on any day is 0.85. The number of messages received may be considered independent from day to day.

(a) Calculate the probability that, during a 16-day period, Janet receives at least one e-mail message on exactly12 days.

(3)

(b) Determine the probability that, during a 30-day period, Janet receives at least one e-mail message on more than 21 days but fewer than 28 days.

(4)

(Total 7 marks)

Q9.

Jenny is practising for a clay-pigeon shooting competition. The probability that she scores a hit with any one shot is 0.8.

Assume that each shot is independent of every other shot.

(a) Write down the probability distribution for Y, the number of shots that Jenny takes at the target in order to score her first hit.

(1)

(b) Calculate the probability that Jenny first hits the target with her fifth attempt.

(2)

(c) Find values for the mean and variance of Y.

(2)

(Total 5 marks)

Q10.

A bin contains 400 coloured erasers that fit on the ends of pencils. The number of erasers of each colour is as follows.

Colour	Green	Blue	Red	Yellow
Number	88	60	160	92

(a) A random sample of 25 erasers is selected, **with replacement**, from the bin. Find the probability that:

(i) exactly 2 erasers are green;

(4)

(ii) at most 3 erasers are blue;

(2)

(iii) between 8 erasers and 12 erasers, inclusive, are red.

(4)

(b) Erasers are selected at random, **without replacement**, from the bin until 5 yellow erasers are obtained.

Give **two** reasons why a binomial distribution does **not** model the number of erasers selected.

(2)

(Total 12 marks)

Q11.

A recent large-scale survey established that 15 per cent of cars have faulty brake lights.

(a) Calculate the probability that, in a random sample of 18 cars, exactly 2 cars have faulty brake lights.

(3)

(b) Determine the probability that, in a random sample of 50 cars, more than 5 cars but fewer than 10 cars have faulty brake lights.

(3)

(c) Find the probability that, in a random sample of 900 cars, at most 150 cars have faulty brake lights.

(5)

(d) At a set of traffic lights, a policewoman records the number, R, of **vehicles** with faulty brake lights, out of 50 successive vehicles stopping at the traffic lights.

Give a reason why the binomial distribution, B(50, 0.15), might **not** be an appropriate model for R.

(1)

(Total 12 marks)

Q12.

The proportions of people with blood groups O, A, B and AB in a particular population are in the ratio 48 : 35 : 12 : 5, respectively.

(a) Determine the probability that a random sample of 20 people from the population contains:

(i) exactly 10 with blood group O;

(4)

(ii) at most 2 with blood group AB;

(2)

(iii) at least 8 with blood group A.

(3)

(b) A random sample of 400 people is selected from the population and the number, R, with blood group B is recorded.

Determine $P(R > 40)$.

(4)

(Total 13 marks)

Q13.

(a) The probability that Azher, who always walks to work, arrives late on any day is 0.11.

Calculate the probability that, during a period of 20 working days, he is late on exactly 4 days.

(3)

(b) The probability that Brenda, who always cycles to work, arrives late on any day is 0.15.

Find the probability that, during a period of 50 working days, she is late on at most 10 days.

(2)

(c) The probability that Chu, who always travels to work by public transport, arrives late on any day is 0.30.

Calculate the probability that, during a period of 120 working days, he is late on more than 30 days but fewer than 40 days.

(7)

(Total 12 marks)

Q14.

Garden canes have lengths that are normally distributed with mean 208.5 cm and standard deviation 2.5 cm.

(a) Show that the probability of the length of a randomly selected cane being between 205 cm and 210 cm is 0.645, correct to three decimal places.

(4)

(b) Ten canes are selected at random.

Calculate the probability that exactly 6 of these canes have lengths between 205 cm and 210 cm.

(3)
(Total 7 marks)

Q15.

Paper clips are produced in a variety of colours.

The proportion of red paper clips produced is 0.20.

Determine the probability that, in a random sample of 50 coloured paper clips, the number of red clips is:

(a) fewer than 10;

(3)

(b) at least 8 but at most 12.

(3)
(Total 6 marks)

Q16.

The probability that at least one computer is available at any time in an 'Internet Cafe' is 0.75.

(a) Mr World makes 16 visits to the cafe. Calculate the probability that, on entry, at least one computer is available on exactly 10 occasions.

(3)

(b) Miss Wyde makes 30 visits to the cafe. Determine the probability that, on entry, at least one computer is available on 20 or more occasions.

(4)
(Total 7 marks)

Q17.

A manufacturer of balloons produces 40 per cent that are oval and 60 per cent that are round.

Packets of 20 balloons may be assumed to contain random samples of balloons. Determine the probability that such a packet contains:

(a) an equal number of oval balloons and round balloons;

(3)

(b) fewer oval balloons than round balloons.

(2)

A customer selects packets of 20 balloons at random from a large consignment until she finds a packet with exactly 12 round balloons.

(c) Give a reason why a binomial distribution is **not** an appropriate model for the number of packets selected.

(1)
(Total 6 marks)

Q18.

Toothbrushes have bristles that are either firm, medium or sensitive. The proportions of toothbrushes with these bristles are 0.12, 0.53 and 0.35 respectively.

A random sample of 50 toothbrushes is selected.

(a) Calculate the probability that the sample contains exactly 6 toothbrushes with **firm** bristles.

(3)

(b) Determine the probability that the sample contains at most 20 toothbrushes with **sensitive** bristles.

(2)
(Total 5 marks)

Q19.

A chemist sells two types of bath cube. One type is relaxing and the other type is invigorating. The owner of a small hotel buys 50 cubes from the chemist. For each cube there is independently a probability of 0.40 that it will be relaxing.

(a) Find the probability that the 50 cubes will include:

(i) 20 or fewer relaxing cubes;

(3)

(ii) 20 or fewer invigorating cubes;

(3)

(iii) more relaxing cubes than invigorating cubes.

(2)

(b) The 50 cubes included 22 relaxing cubes. State, giving a reason, whether or not a binomial distribution will provide an appropriate model for the random variable, R, in **each** of the following cases.

(i) A guest at the hotel randomly selects one of the 50 cubes. If it is not a relaxing cube, he replaces it and again selects a cube at random. He continues this procedure until he obtains a relaxing cube. The random variable R denotes the number of cubes he selects until he obtains a relaxing cube.

(ii) The owner randomly selects 20 of the 50 cubes and places them in a bowl.

The random variable R denotes the number of relaxing cubes placed in the bowl.

(4)
(Total 12 marks)

Q20.

Eight friends take a picnic to a cricket match. As her contribution to the picnic, Hilda buys eight sandwiches at a supermarket. She selects the sandwiches at random from those on display. The probability that a sandwich is suitable for vegetarians is independently 0.3 for each sandwich.

(a) Find the probability that, of the eight sandwiches, the number suitable for vegetarians is:

(i) 2 or fewer;

(ii) exactly 2;

(iii) more than 3.

(7)

(b) Two of the eight friends are vegetarians. Hilda decides to ensure that the eight sandwiches she takes to the match will include at least two suitable for vegetarians. If, having selected eight sandwiches at random, she finds they include fewer than two suitable for vegetarians she will replace one, or if necessary two, of the sandwiches unsuitable for vegetarians with the appropriate number of sandwiches suitable for vegetarians.

State whether or not the binomial distribution provides an appropriate model for the number of sandwiches suitable for vegetarians which Hilda takes to the match. Explain your answer.

(2)

(c) In fact the eight sandwiches which Hilda took to the match contained four suitable for vegetarians. The first four friends to eat a sandwich were not vegetarians. Each selected one of the available sandwiches at random and ate it.

State whether or not the binomial distribution provides an appropriate model for the number of sandwiches suitable for vegetarians eaten by these four friends. Explain your answer.

(2)
(Total 11 marks)

Q21.

Gerhard walks to school each morning. The probability that he arrives late is 0.15 and is independent of whether he arrives late on any other morning.

(a) Find, for a week when he walks to school on five mornings:

(i) the probability he arrives late on two or fewer mornings;

(3)

(ii) the probability he arrives late on more than three mornings;

(2)

(iii) the mean and the standard deviation of the number of mornings on which he arrives late.

(3)

(b) The following table summarises the number of late arrivals of all pupils who attended Gerhard's school on five mornings of a particular week.

Number of late arrivals	Number of pupils
0	275
1	111
2	33
3	12
4	13
5	16

(i) Calculate the mean and the standard deviation of the data in the table.

(3)

(ii) Show that, for the pupils represented in the table, an estimate of the probability of a pupil arriving late on a particular morning is 0.15.

(1)

(c) It is suggested that the data in the table could be modelled by a binomial distribution.

(i) Comment on this suggestion in the light of your calculations in parts (a) and (b).

(2)

(ii) In the context of this question, give **two** possible reasons why the binomial distribution may **not** be a suitable model for the data in part (b).

(2)

(Total 16 marks)

Q22.

Spyros manages a unit trust. He invests clients' money in companies listed on the stock market and is judged each year on whether or not his investments have outperformed the stock market average.

Spyros decides which companies to invest in by asking his young daughter to stick a pin in a list of possible companies. Using this strategy, the probability that he will outperform the stock market average in any particular year is 0.5. This probability may be assumed independent for each year.

(a) Find the probability that, in a 6-year period, Spyros will outperform the stock market average in:

(i) more than 4 years;

(3)

(ii) all 6 years.

(2)

(b) In 1997, a journal reported that out of 900 unit trusts only 14 had managed to outperform the stock market average in each of the 6 years from 1991 to 1996. Comment on this in the light of your answer to part (a)(ii).

(3)
(Total 8 marks)

Q23.

An airline operates a service between Manchester and Paris. The flight time may be modelled by a normal distribution with mean 85 minutes and standard deviation 8 minutes.

(a) Find the probability that a particular flight time is:

(i) less than 75 minutes;

(3)

(ii) between 75 minutes and 81 minutes.

(3)

(b) In order to gain publicity for the service, the airline decides to refund fares when a flight time exceeds q minutes. Find the value of q such that the probability of fares being refunded on a particular flight is 0.001.

(4)

(c) Find the probability that the mean of four randomly selected flight times will be less than 81 minutes.

(4)
(Total 14 marks)

Q24.

An airline company accepts bookings for 20 executive-class seats on a flight from London to Bergen. The probability that a customer, who has booked an executive-class seat, will not turn up to claim the seat is 0.15 and may be assumed to be independent of the behaviour of other customers.

(a) Find the probability that, of the customers who have booked executive-class seats:

(i) all will turn up to claim their seats;

(ii) one or more will **not** turn up to claim their seat.

(4)

(b) The airline also accepts bookings for 50 standard-class seats on the same flight. The probability that a customer, who has booked a standard-class seat, will not turn up to claim the seat is 0.08 and may be assumed to be independent of the behaviour of other customers.

Find the probability that, of the customers who have booked standard-class seats:

(i) 2 or more will **not** turn up;

(ii) 3 or more will **not** turn up

(4)

(c) The aeroplane has 19 executive-class seats and 48 standard-class seats. If 19 or fewer customers who have booked executive-class seats turn up they will all be allocated executive-class seats. If all 20 customers who have booked executive-class seats turn up, one will be allocated a standard-class seat. This will leave only 47 seats available for the 50 customers who have booked standard-class seats.

Use your results from parts (a) and (b) to calculate the probability that there will be standard-class seats available for all those who turn up having booked standard-class seats.

(5)
(Total 13 marks)

Q25.

A gardener plants beetroot seeds. The probability of a seed not germinating is 0.35, independently for each seed.

Find the probability that, in a row of 40 seeds, the number not germinating is:

(a) 9 or fewer;

(3)

(b) 7 or more;

(2)

(c) equal to the number germinating.

(3)
(Total 8 marks)

Q26.

Each house in a local authority area is supplied with a recycling box for discarded bottles. The council arranges a fortnightly collection of the boxes from outside the houses.

In a street containing 30 houses the number of boxes placed outside for a particular collection is a random variable X. The values of X for 9 consecutive collections were:

15 12 10 11 12 14 11 10 13

(a) Calculate the mean and standard deviation of the given data.

(2)

(b) Use the data to estimate p, the proportion of houses that a box will be placed outside for a particular collection.

(1)

(c) Suggest the name of a distribution which might provide a suitable model for the random variable X.

(1)

(d) Use the distribution you have suggested in part (c) and the value of p you have estimated in part (b) to find:

(i) the probability that X exceeds 15;

(3)

(ii) the mean and standard deviation of X.

(3)

(e) By comparing your answers to part (a) with your answers to part (d)(ii), explain whether the distribution you have suggested in part (c) does provide a suitable model for X.

(2)

(f) Regardless of your calculations and of your answer to part (e), give a reason why the distribution you have suggested in part (c) may not provide a suitable model for X.

(1)
(Total 13 marks)

Q27.

Jeremy sells a magazine which is produced in order to raise money for homeless people. The probability of making a sale is, independently, 0.09 for each person he approaches. Given that he approaches 40 people, find the probability that he will make:

(a) 2 or fewer sales;

(3)

(b) exactly 4 sales;

(2)

(c) more than 5 sales.

(2)
(Total 7 marks)

Q28.

Siballi is a student who travels to and from university by bus. He believes that the probability of having to wait more than 6 minutes to catch a bus is 0.4 and is independent of the time of day and direction of travel.

(a) Assuming that Siballi's beliefs are correct, find the probability that, during a particular week when he catches 10 buses, the number of times he has to wait more than 6 minutes is:

(i) 3 or fewer;

(3)

(ii) more than 4.

(2)

(b) Assuming that Siballi's beliefs are correct, calculate values for the mean and standard deviation of the number of times he has to wait more than six minutes to catch a bus in a week when he catches 10 buses.

(3)

(c) During a thirteen-week period, the numbers of times (out of 10) he had to wait more than six minutes to catch a bus were as follows:

4	8	8	9	3	2	2	7	0	1	5	2	0

(i) Calculate the mean and standard deviation of these data.

(2)

(ii) State, giving reasons, whether your answers to part (c)(i) support Siballi's beliefs that the probability of having to wait more than 6 minutes to catch a bus is 0.4 and is independent of the time of day and direction of travel.

(3)

(Total 13 marks)

Q29.

A specialist bicycle shop builds made-to-measure bicycle frames. The shop has 40 orders for frames. Past experience shows that the probability of a customer, who has ordered a frame, failing to complete the purchase is 0.04 and is independent for each customer.

(a) Find the probability that of the 40 orders, the number of purchases not completed is:

(i) 4 or fewer;

(ii) exactly 2.

(6)

(b) Find the probability that the purchase is completed for all 40 orders.

(2)

(Total 8 marks)